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Summary

- Python developer in Berlin. Currently learning the Python backend framework Django!
Currently developing software in python as a Machine learning engineer in Berlin, proficient in data analysis. I enjoy hard problems that require my quantitative, programming and analytical skills developed during my master's degree in mathematics. Often I find that careful data cleaning and data analysis provide insights into data that help me train better models, and basic statistics and probability provide excellent sanity checks. I have frequently worked with time-series data.

Employment History

1st August 2022 – 31st December 2022

- Machine Learning Engineer (Intern). Smart Steel Technologies GmbH, Berlin, Germany.

Summary

- German job-seeking visa already with permission to work full time : No visa sponsorship necessary.
- Earliest Entry Date: 01.01.2023.
- Salary Expectations: € 48000-50000 per annum.

Education

2022 M.Sc. Mathematics, Georg-August-Universität Göttingen, Germany.
Thesis title: *The Möbius Function on Arithmetic Progressions.*

2017 B.S (Research). Mathematics, Shiv Nadar University, India.
Thesis title: *Dirichlet's Theorem on Arithmetic Progressions.*

Skills

Technical Knowledge	Supervised and Unsupervised Machine learning, clustering, k -nearest neighbours algorithm, Mathematics, Probability and Statistics, Coding, Linux command line, Git : Gitlab and Github.
Languages	English, Hindi, Bengali – Fluent, German – A2.
Coding	Python, SQL, $\text{\LaTeX}$.
Python packages	Scikit-learn, Pandas, XGBoost , Seaborn, Matplotlib, SciPy, Numpy, Pytest, Jupyter, Docker.
Relational Databases	PostgreSQL, Clickhouse
Misc.	Academic research, teaching.

Contributions

- Built a proof of concept machine learning model on imbalanced tabular dataset that predicts defects in steel post-casting, beating random forests and XGBoost in terms of recall with similar precision. This classification model was rule-based, informed by insights supplied by deeper analysis of the data set.
- Used libraries like pandas (matplotlib backend) and Seaborn to detect anomolous behaviour, thereby quantifying damage caused by a certain leakage in customer's steel plant.

Responsibilities

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|------------------|---|---|
| Data Cleaning | ■ | Using pandas to clean real industrial time-series data from steel plants. |
| Visualisation | ■ | Using Matplotlib and Seaborn to obtain insights into data |
| Tests | ■ | Writing Pytests to test code written by colleagues. |
| Machine Learning | ■ | Building proof-of-concept machine learning models that predict defects in certain grades of steel |

Miscellaneous Experience

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|---------|---|--|
| Udemy | ■ | Course: Machine Learning, Data Science and Deep Learning with Python. Link to Certificate. |
| Hobbies | ■ | Floorball and badminton, mathematical research. |